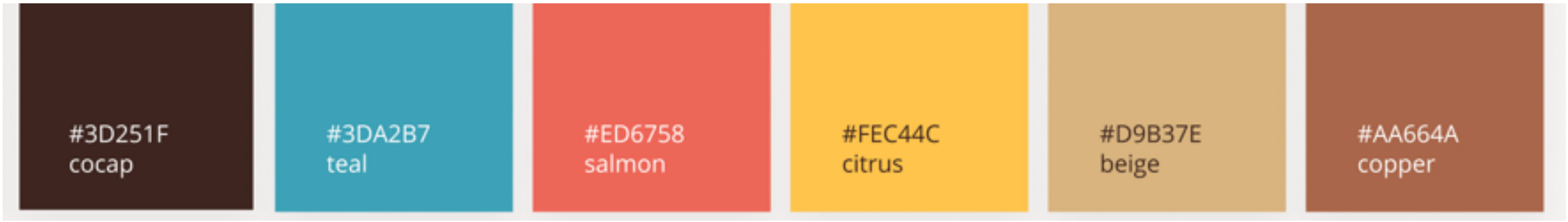


Theorem [Brenier, 1991]:

Assumptions: $c(x, y) = \|x - y\|^2$, α has a density $\rho_\alpha = \frac{d\alpha}{dx}$

Then:

- Kantorovich problem has a **unique** solution π^*
- π^* is deterministic, supported on $(x, T(x))$
- Where T is the gradient of a convex function



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

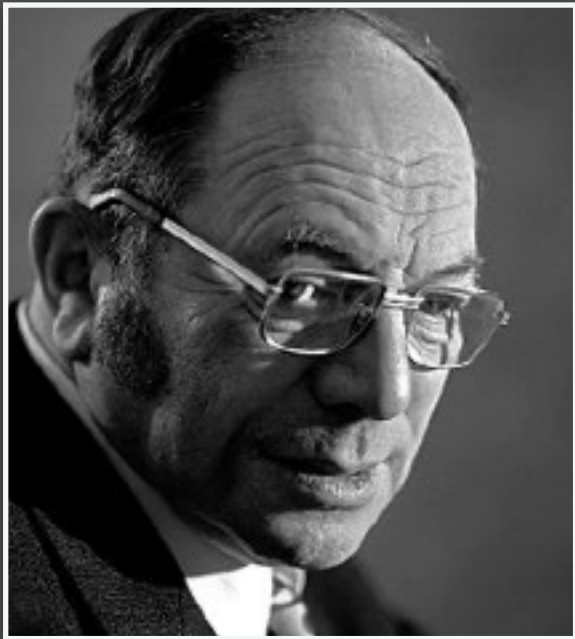
#AA664A
copper



Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m C_{i,j} \pi_{i,j}$$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$

Theorem [Birkhoff-von Neumann]:

Every bistochastic matrix can be written as a convex combination of permutation matrices:

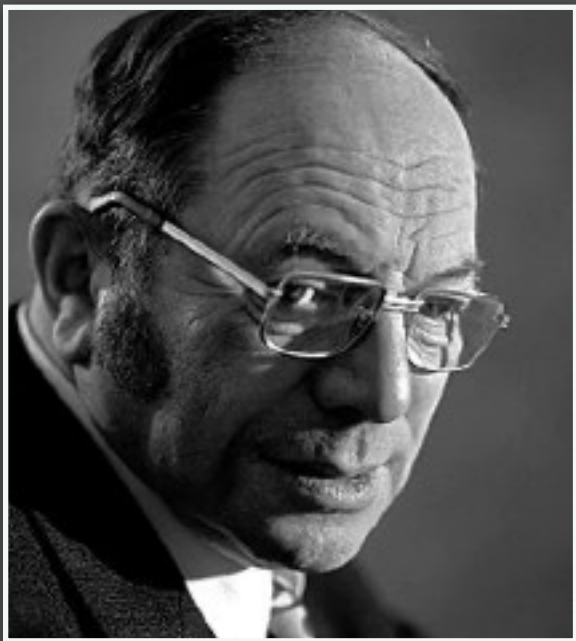
$$M = \sum_{k=1}^K \lambda_k P_k, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1.$$

Key Takeaway

Couplings generalize maps:

Monge $\subset$ Kantorovich. This relaxation guarantees existence and convexity.

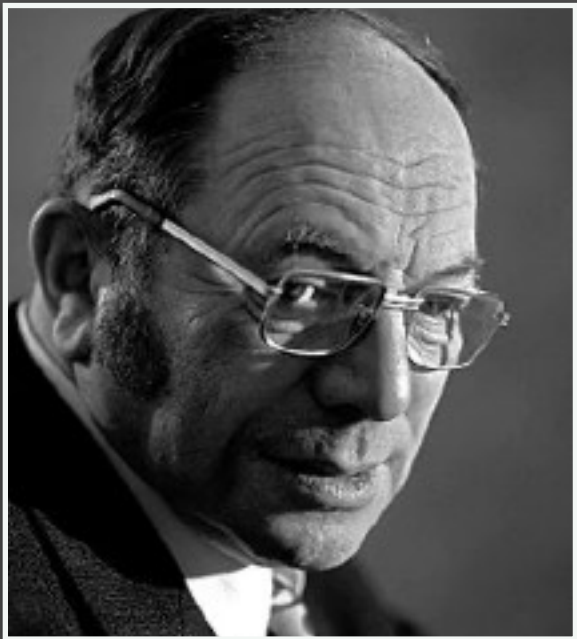




Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m C_{i,j} \pi_{i,j}$$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$